/**

* hive exercises

*

* part of a longer hadoop video course at http://bit.ly/learn-hadoop/

*

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*/

// our sample "page_view" data looks like this:

```
1391102617,1234,http://quicloud.com/,http://google.com/,198
.211.110.9

1391101111,1234,http://somewhere.com/,http://google.com/,19
7.211.110.9

1391002637,1234,http://somwhere-
else.com/,http://yahoo.com/,196.211.110.9

1391002617,1234,http://quicloud.com/,http://bing.com/,198.2
11.110.9
```

// start up hive

```
hive
```

// this is an external table

```
CREATE DATABASE page_view;
```

// note: if using the AWS-backed 'cloudera manager' stack, change '/user/cloudera/' to '/user/ubuntu/'

use page_view;

```
CREATE EXTERNAL TABLE page_view_external(viewTime INT,
userid BIGINT,page_url STRING, referrer_url STRING,ip
STRING) COMMENT 'This is the EXTERNAL page view table' ROW
FORMAT DELIMITED FIELDS TERMINATED BY ',' LOCATION
'/user/input-data/page_view/';
```

```
DESC page_view_external;

DESC EXTENDED page_view_external;
```

//notice tableType (EXTERNAL_TABLE)

// this command returns quickly (probably seconds). why?

// ...if we are not doing any filtering (like where), and returning all rows, no MR is done

```
SELECT * FROM page_view_external;
```

// this command, however, takes several seconds

```
SELECT * FROM page_view_external WHERE page_url LIKE
'%quicloud%';

SELECT DISTINCT(page_url) FROM page_view_external WHERE
page_url LIKE '%quicloud%';
```

// this is an internal table

```
CREATE TABLE page_view_internal(viewTime INT, userid
BIGINT,page_url STRING, referrer_url STRING,ip STRING)
COMMENT 'This is the INTERNAL page view table' ROW FORMAT
DELIMITED FIELDS TERMINATED BY ',';

LOAD DATA LOCAL INPATH '/home/code-and-data/data/page_view'
INTO TABLE page_view_internal;
```

// What happened to our "local" data? In another terminal do:

```
ls -lat ~/code-and-data/data/page_view/
```

// still there

// validate that the data made it into hive

```
SELECT * FROM page_view_internal;
```

// in another terminal, let's look for where the data lives

```
hadoop fs -ls /user/hive/warehouse/page_view_internal

hdfs dfs
```

```
start-all.sh
```

```
start-dfs.sh
```

```
start-yarn.sh
```

// where does our "external" (page_view_external) tables data live? this should work, right?

```
hadoop fs -ls /user/hive/warehouse/
```

// WRONG! This data is EXTERNAL and in the location that we connected up with the CREATE TABLE command

```
hadoop fs -ls /user/cloudera/input-data/page_view/
```

```
DESC EXTENDED page_view_internal;
```

// notice tableType (MANAGED_TABLE)

// you may want to query the hive data more, but when you're done, let's drop each table & notice what happens

```
SHOW TABLES;
```

```
DROP TABLE page_view_internal;
```

```
DROP TABLE page_view_external;
```

```
SHOW TABLES;
```

// in a 2nd terminal, run these commands. Which data do we expect to persist?

```
hadoop fs -ls /user/hive/warehouse/page_view_internal/
```

```
hadoop fs -ls /user/cloudera/input-data/page_view
```

```
+-----------------------------+-----------------------------+-----------------------------+-----------------------------------+-----------------------+
| page_view_internal.viewtime | page_view_internal.userid   | page_view_internal.page_url | page_view_internal.referrer_url   | page_view_internal.ip |
+-----------------------------+-----------------------------+-----------------------------+-----------------------------------+-----------------------+
| 1391102617                  | 1234                        | http://quicloud.com/        | http://google.com/                | 198.211.110.9         |
| 1391101111                  | 1234                        | http://somewhere.com/       | http://google.com/                | 197.211.110.9         |
| 1391002637                  | 1234                        | http://somwhere-else.com/   | http://yahoo.com/                 | 196.211.110.9         |
| 1391002617                  | 1234                        | http://quicloud.com/        | http://bing.com/                  | 198.211.110.9         |
+-----------------------------+-----------------------------+-----------------------------+-----------------------------------+-----------------------+
4 rows selected (2.915 seconds)
0: jdbc:hive2://> select distinct page_url from page_view_internal;
24/10/16 00:27:46 [45ad8965-d3cb-4d2b-bf85-cc270b9ffeca main]: WARN calcite.RelOptHiveTable: No Stats for page_view@page_view_internal, Columns: page_url
No Stats for page_view@page_view_internal, Columns: page_url
24/10/16 00:27:46 [HiveServer2-Background-Pool: Thread-69]: WARN ql.Driver: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions.
Consider using a different execution engine (i.e. tez) or using Hive 1.X releases.
Query ID = hadoopminhchau_20241016002746_ac82786e-9587-45b1-91e4-4672eba58cad
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
WARN  : Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. tez) or using Hive
 1.X releases.
24/10/16 00:27:48 [HiveServer2-Background-Pool: Thread-69]: WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the
Tool interface and execute your application with ToolRunner to remedy this.
Starting Job = job_1729038270886_0001, Tracking URL = http://minhchau-master:9004/proxy/application_1729038270886_0001/
Kill Command = /home/hadoopminhchau/hadoop/bin/mapred job  -kill job_1729038270886_0001
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
24/10/16 00:28:03 [HiveServer2-Background-Pool: Thread-69]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.h
adoop.mapreduce.TaskCounter instead
2024-10-16 00:28:03,775 Stage-1 map = 0%,  reduce = 0%
2024-10-16 00:28:11,077 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 3.56 sec
2024-10-16 00:28:16,273 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 5.09 sec
MapReduce Total cumulative CPU time: 5 seconds 90 msec
Ended Job = job_1729038270886_0001
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 5.09 sec   HDFS Read: 15653 HDFS Write: 192 HDFS EC Read: 0 SUCCESS
Total MapReduce CPU Time Spent: 5 seconds 90 msec
+----------------------------+
|          page_url          |
+----------------------------+
| http://quicloud.com/       |
| http://somewhere.com/      |
| http://somwhere-else.com/  |
+----------------------------+
3 rows selected (31.96 seconds)
0: jdbc:hive2://> _
```

```
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 5.09 sec   HDFS Read: 15653 HDFS Write: 192 HDFS EC Read: 0 SUCCESS
Total MapReduce CPU Time Spent: 5 seconds 90 msec
+----------------------------+
|          page_url          |
+----------------------------+
| http://quicloud.com/       |
| http://somewhere.com/      |
| http://somwhere-else.com/  |
+----------------------------+
3 rows selected (31.96 seconds)
0: jdbc:hive2://> select distinct page_url from page_view_internal where page_url like '%quicloud%';
24/10/16 00:30:44 [45ad8965-d3cb-4d2b-bf85-cc270b9ffeca main]: WARN calcite.RelOptHiveTable: No Stats for page_view@page_view_internal, Columns: page_url
No Stats for page_view@page_view_internal, Columns: page_url
24/10/16 00:30:45 [HiveServer2-Background-Pool: Thread-95]: WARN ql.Driver: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions.
Consider using a different execution engine (i.e. tez) or using Hive 1.X releases.
Query ID = hadoopminhchau_20241016003044_8ce3f21c-c29a-48a0-9069-4e16a40a7cf0
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
24/10/16 00:30:46 [HiveServer2-Background-Pool: Thread-95]: WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the
Tool interface and execute your application with ToolRunner to remedy this.
WARN  : Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. tez) or using Hive
 1.X releases.
Starting Job = job_1729038270886_0002, Tracking URL = http://minhchau-master:9004/proxy/application_1729038270886_0002/
Kill Command = /home/hadoopminhchau/hadoop/bin/mapred job  -kill job_1729038270886_0002
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
24/10/16 00:30:52 [HiveServer2-Background-Pool: Thread-95]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.h
adoop.mapreduce.TaskCounter instead
2024-10-16 00:30:52,931 Stage-1 map = 0%,  reduce = 0%
2024-10-16 00:30:59,136 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 3.07 sec
2024-10-16 00:31:04,313 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 4.55 sec
MapReduce Total cumulative CPU time: 4 seconds 550 msec
Ended Job = job_1729038270886_0002
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 4.55 sec   HDFS Read: 16299 HDFS Write: 120 HDFS EC Read: 0 SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 550 msec
+----------------------+
|       page_url       |
+----------------------+
| http://quicloud.com/ |
+----------------------+
1 row selected (20.648 seconds)
0: jdbc:hive2://> _
```